

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Historical trajectory 1750–2024 — who emits, how much, and why it matters

1750–2024 Scope

35+ Bt Annual Emissions

Per Capita Inequalities

Atmospheric CO₂ Reaches 429 ppm

— 50% Above Pre-Industrial Levels

429

parts per million (ppm)

Mauna Loa Observatory, Feb 2026

Pre-industrial baseline:

~280 ppm

Current level:

429 ppm (+53%)

Change since last ice age:

Ice-age shift took 10,000+ yrs

Today's rise: ~2 centuries

TIMELINE CONTEXT



Pre-1750

~280 ppm



1958 (Keeling)

315 ppm



2000

370 ppm



2013

400 ppm
(first breach)



Feb 2026

429 ppm

Trajectory

No peak yet

Ground-based and satellite measurements confirm sharp acceleration linked primarily to fossil fuel combustion.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion → 35+ Billion Tonnes

1950

~6 Bt

Slow early growth
post-WWII baseline

1990

~20 Bt

Nearly 4× the 1950 level
post-industrialization

2024

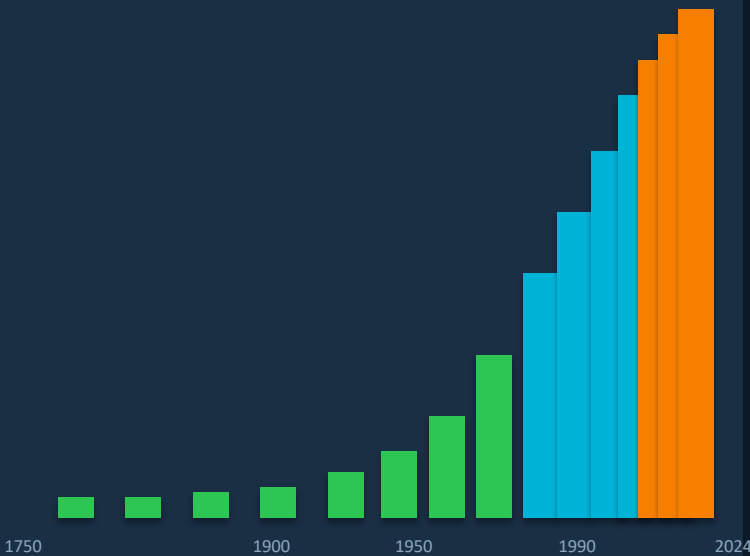
35+ Bt

Still rising
no peak in sight

KEY CONTEXT

- Before the Industrial Revolution, CO₂ emissions were negligible.
- Slow growth until mid-20th century; then rapid acceleration driven by industrialisation and fossil fuels.
- Growth has slowed in recent years — but emissions have not yet peaked.

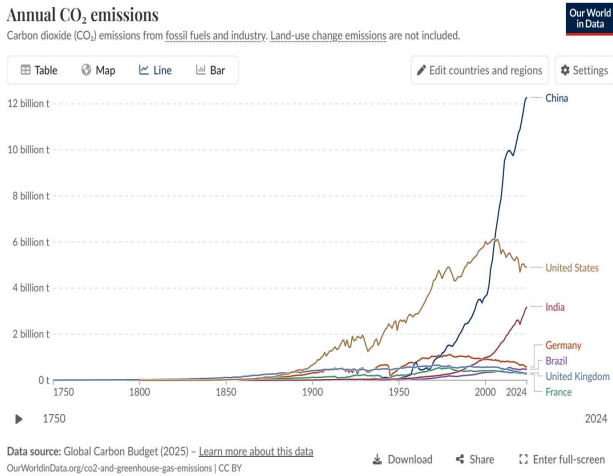
EMISSION GROWTH CURVE (SCHEMATIC)



China Now Emits Over 12 Billion Tonnes — More Than Double the United States

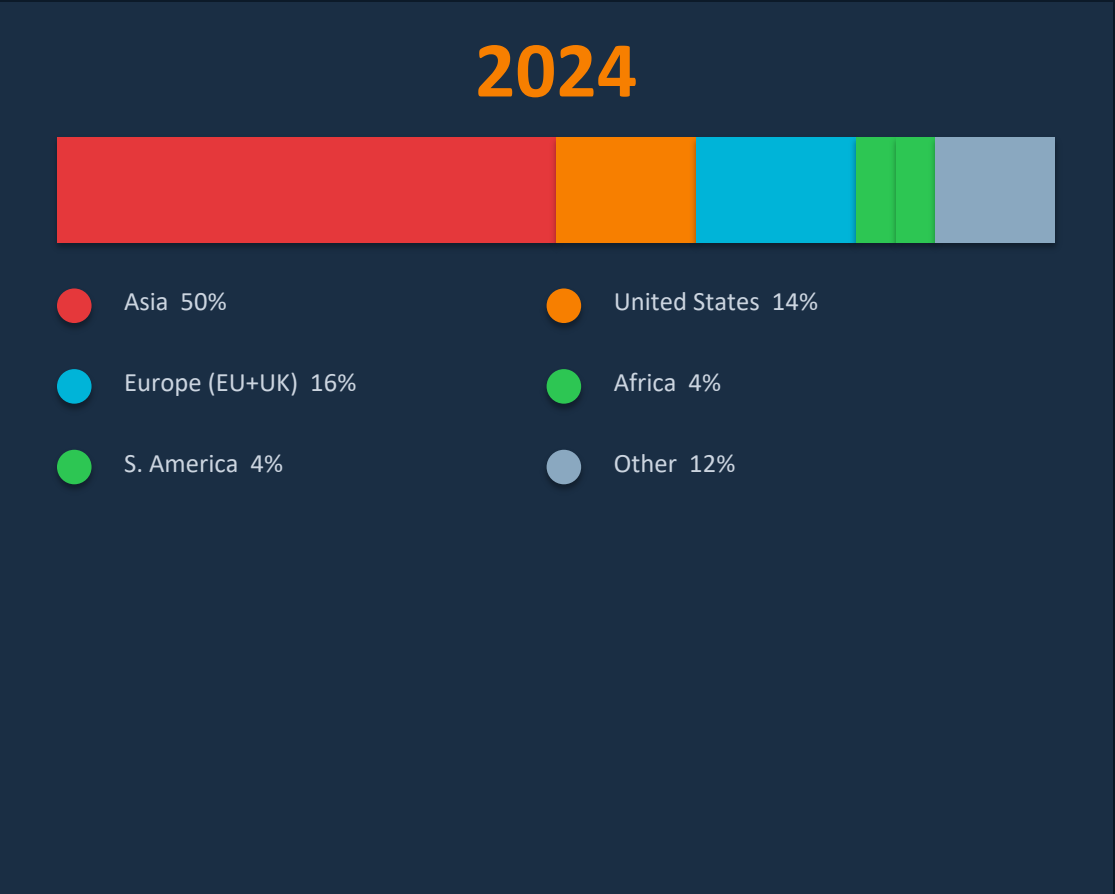
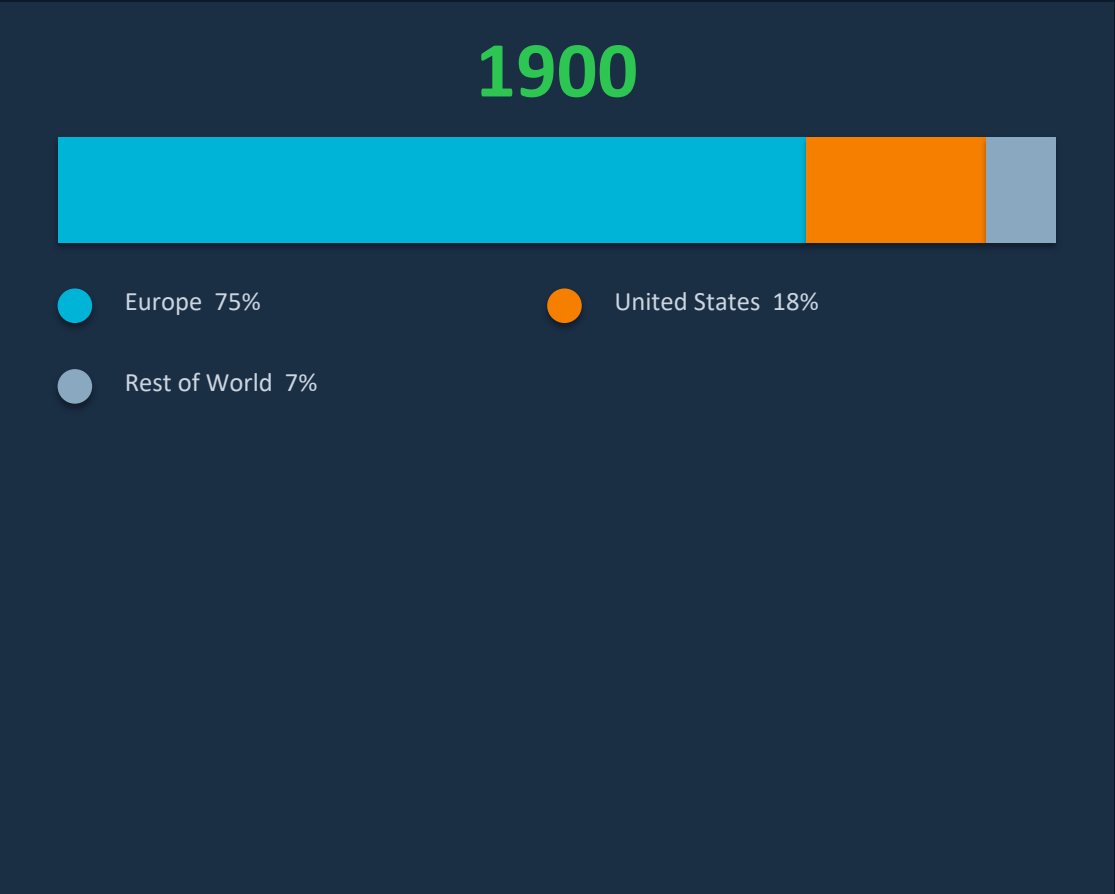
Country / Region	CO ₂ (Bt/yr)	Global %	Relative Scale	
China	~12.5	~27.0%		Slight increase
United States	~5.0	~14.0%		Declining
India	~3.0	~8.0%		Rising
EU-27	~2.7	~7.0%		Declining
Germany	~0.6	~1.7%		Declining
United Kingdom	~0.3	~0.9%		Declining
Brazil	~0.5	~1.4%		Roughly stable
France	~0.3	~0.8%		Declining

figure_3.jpg — ANNUAL CO₂ BY COUNTRY



Key insight:
China alone accounts for 27% of global emissions. The US has declined from a ~6 Bt peak (~2005) to ~5 Bt today, while India continues to rise.

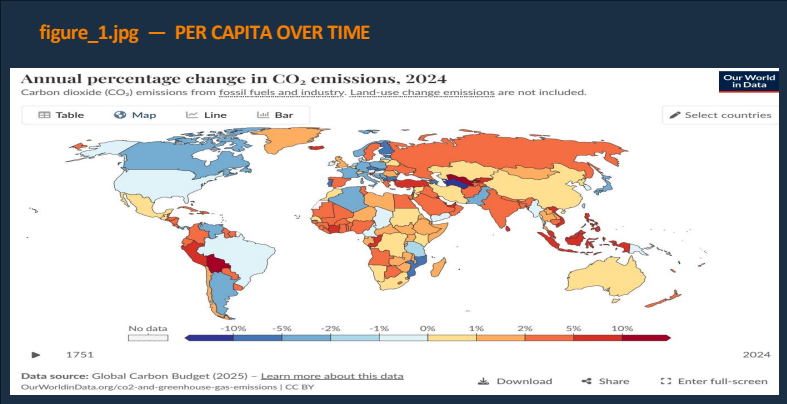
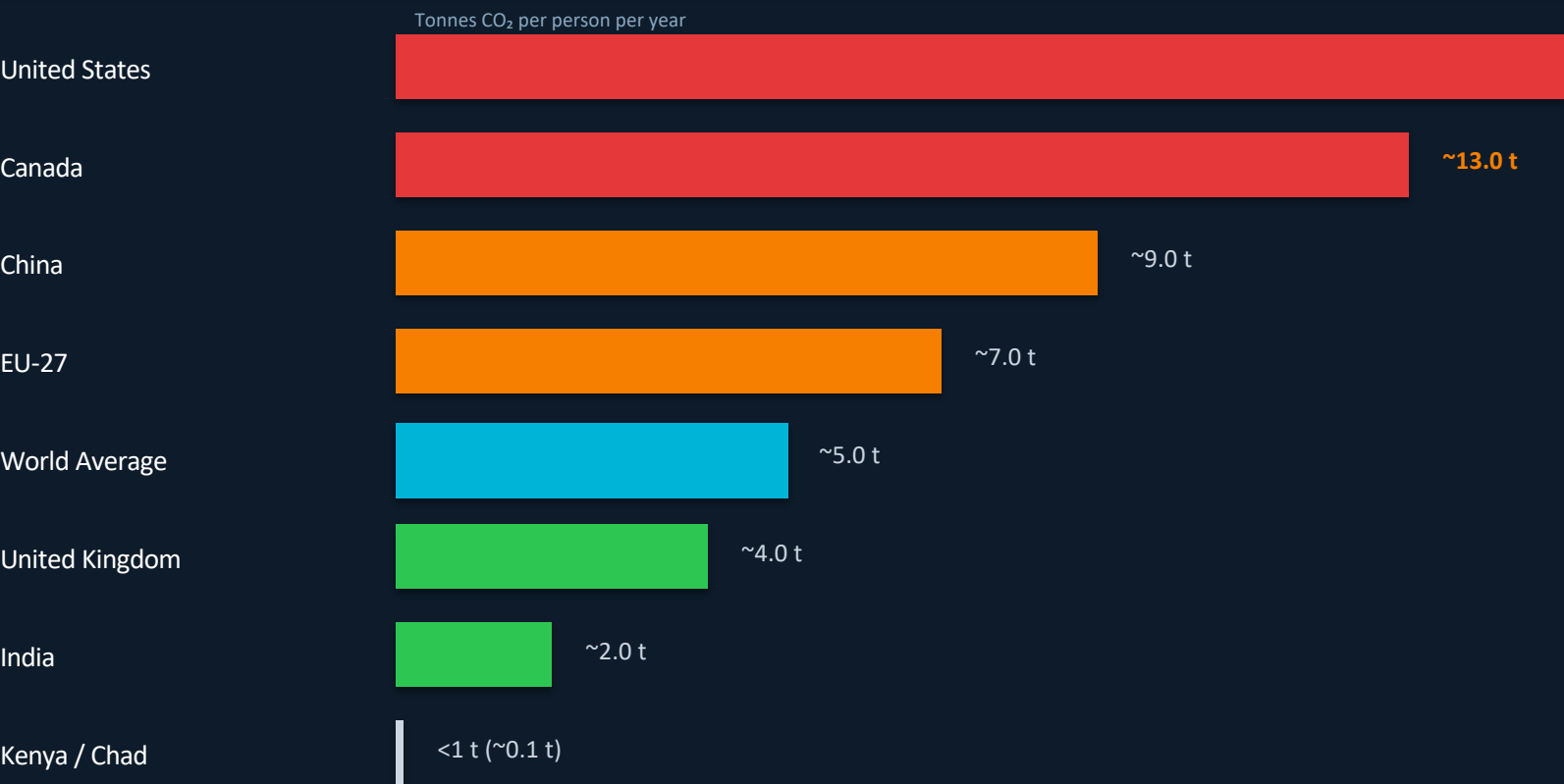
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



Key shift:

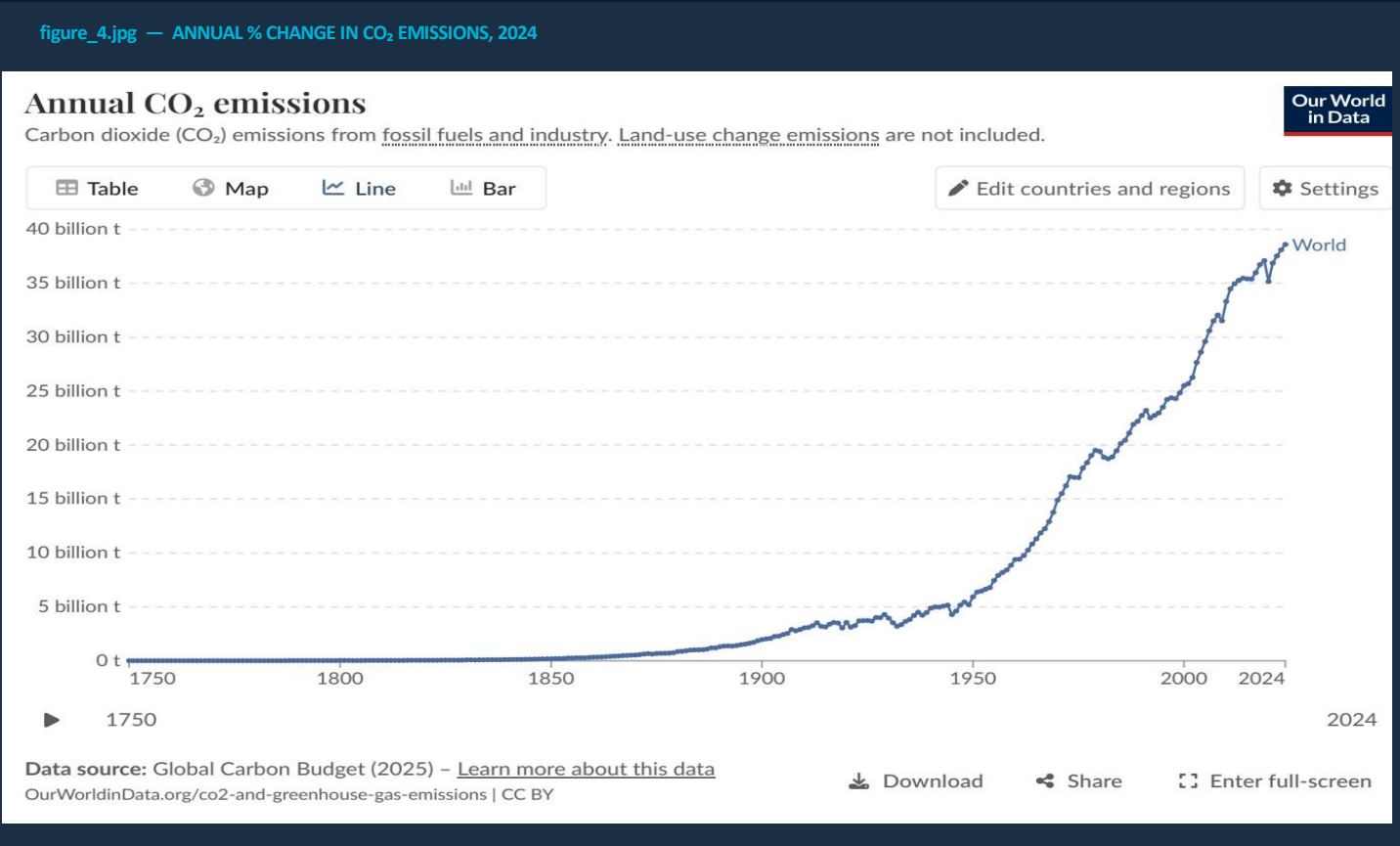
In 1900, Europe + US >90% of global emissions. By 2024: Asia ~50%, US + Europe < 33%. Africa and South America each ~3–4%, comparable to international aviation and shipping.

Per Capita Gap: 15 t (US) vs. 0.1 t (Chad) — a 150× Disparity



The average American emits in under 2 days what the average person in Chad emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



- Strong decrease (-5% to -10%)
- Moderate decrease (-1% to -5%)
- Near zero ($\pm 1\%$)
- Moderate increase (+2% to +10%)
- Strong increase ($>+10\%$)

REGIONAL HIGHLIGHTS

US + Most of Europe	-1% to -5%
Russia	+2% to +5%
Parts of SE Asia + Africa	+5% to +10%+
Some S. America nations	>+10%
Others in S. America	Declines

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar prosperity and living standards can have very different per capita CO₂ emissions depending on their reliance on nuclear, renewables, or fossil fuels.

Country	Nuclear Share	Renewables Share	Fossil Fuel Reliance	Per Capita CO ₂	
France	~70% nuclear	~25% renewables	Low fossil reliance	~4 t/person	LOW EMITTER
United Kingdom	~15% nuclear	~40% renewables	Moderate, declining	~4 t/person	LOW EMITTER
Germany	~0% nuclear (phased out)	~46% renewables	Higher coal/gas share	~8 t/person	HIGH EMITTER
Netherlands	~3% nuclear	~35% renewables	High natural gas reliance	~9 t/person	HIGH EMITTER
KEY INSIGHT Policy and energy mix choices — not just economic prosperity — determine per capita CO₂ outcomes. France and the UK have kept emissions around 4 t/person through high nuclear and renewables shares, while Germany and the Netherlands remain above 8–9 t/person due to greater fossil fuel dependence.					

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

01

Global emissions exceed 35 Bt and have not yet peaked

Global annual CO₂ from fossil fuels and industry exceeds 35 billion tonnes. Growth has slowed but emissions continue to rise — no peak has been reached.

02

China alone emits >25% of the global total

China produces ~12.5 Bt/yr (~27%). India (~3 Bt) is the third-largest emitter and rising rapidly. The US has declined from its ~2005 peak to ~5 Bt.

03

Per capita inequalities remain extreme — a 150× gap

The US and Australia emit ~15 t/person; Chad and Niger ~0.1 t/person. The average American emits in under 2 days what the average person in Chad emits in a year.

04

US and Europe declining, but historical cumulative responsibility remains large

The geographic centre of emissions has shifted: from >90% Europe + US in 1900 to Asia ~50% today, while US + Europe account for less than one-third.

05

Policy and energy mix — not just prosperity — determine per capita outcomes

France (4 t) vs. Germany (8 t): similar wealth, very different emissions. Nuclear and renewable energy choices matter enormously.